

Polynomials

Aasan Hinglish Notes, Examples aur Step-by-Step Solutions

Polynomial ek algebraic expression hai jisme variables aur constants hote hain. Is guide me variable/constant se lekar degree, coefficient, zeros, factorisation aur graph tak sab kuch simple Hinglish me samjhaya gaya hai - solved examples aur 20 practice questions ke step-by-step solutions ke saath.

EXAM * aise green tag wale topics exams me baar-baar aate hain

1. Variable aur Constant

EXAM * Exam me CONSTANTLY aata hai

Variable: Badalne wala (unknown) number, jaise x , y , z .

Constant: Fixed number jiski value nahi badalti, jaise 5 , -3 , $1/2$.

- Example: $4x + 7$ me ' x ' variable hai aur ' 7 ' constant hai.

2. Power / Exponent

EXAM * Exam me CONSTANTLY aata hai

Matlab: Power batata hai ki number ko kitni baar khud se multiply karna hai.

- $x^2 = x$ times x (ise ' x square' bolte hain).
- $x^3 = x$ times x times x (ise ' x cube' bolte hain).
- Koi bhi number ki power 0 ho to value 1 hoti hai ($x^0 = 1$).

3. Polynomial Kya Hai?

EXAM * Exam me CONSTANTLY aata hai

Definition: Aisa expression jisme variable ke power SIRF whole numbers ($0, 1, 2, 3, \dots$) hon - negative ya fraction power nahi.

Polynomial Hai $2x + 3$, $x^2 - 5x + 6$, 7

Polynomial Nahin $x^{-1} + 2$, $\sqrt{x} + 1$ (in me galat power hai)

4. Degree

EXAM * Exam me CONSTANTLY aata hai

Definition: Polynomial ka sabse bada power hi uska degree hota hai.

- $2x + 5 \rightarrow$ degree 1
- $x^2 - 3x + 1 \rightarrow$ degree 2
- $5x^3 - 2x \rightarrow$ degree 3

5. Types (Degree ke Hisaab se)

- Degree $0 =$ CONSTANT polynomial (jaise 7)
- Degree $1 =$ LINEAR : $ax + b$
- Degree $2 =$ QUADRATIC : $ax^2 + bx + c$
- Degree $3 =$ CUBIC : $ax^3 + bx^2 + cx + d$
- (In sab me ' a ' hamesha $\neq 0$ hona chahiye.)

Terms ke hisaab se:

- Monomial = 1 term ($3x$), Binomial = 2 terms ($x + 2$), Trinomial = 3 terms ($x^2 + 2x + 1$).

6. Coefficient

EXAM * Exam me CONSTANTLY aata hai

Definition: Variable ke aage wala number hi coefficient hota hai.

Example: $3x^2 - 5x + 7$ me:

- x^2 ka coefficient (a) = 3
- x ka coefficient (b) = -5
- constant (c) = 7

7. $p(x)$ Notation aur Substitution

$p(x)$: Polynomial ka naam (jaise function).

$p(2)$: x ki jagah 2 daal kar value nikalna.

- Example: $p(x) = x^2 - 4 \rightarrow p(2) = 2^2 - 4 = 0$.

8. Zero / Root of Polynomial

EXAM * Exam me CONSTANTLY aata hai

Definition: Wo value of x jiske liye $p(x) = 0$ ho jaye, usi ko polynomial ka ZERO (ya root) kehte hain.

- Example: $p(x) = 2x - 6 \rightarrow 2x - 6 = 0 \rightarrow x = 3$. So zero = 3.
- Graph jahan x -axis ko cut kare, wahi zero hota hai.
- Linear ka 1 zero, Quadratic ke 2 zeros, Cubic ke 3 zeros (maximum).

9. Factorisation (Splitting the Middle Term)

EXAM * Exam me CONSTANTLY aata hai

Idea: $ax^2 + bx + c$ ko $(..)(..)$ ke product me todna.

- Middle term 'b' ko aise 2 numbers p, q me todo jahan:
- $p + q = b$ AND $p \times q = a \times c$.
- Example: $x^2 + 5x + 6 \rightarrow p+q=5, pq=6 \rightarrow 2,3 \rightarrow (x+2)(x+3)$.

10. Graph Basics

EXAM * Exam me CONSTANTLY aata hai

- X -axis = leti (horizontal) line, Y -axis = khadi (vertical) line.
- Linear polynomial ka graph = seedhi line.
- Quadratic ka graph = 'U' ya ulta 'U' (ise PARABOLA kehte hain).
- Graph jahan x -axis ko touch/cut kare = wahi zero hota hai.

Quick Revision

Degree = sabse bada power.

Coefficient = variable ke aage ka number.

Zero = jahan $p(x) = 0$ (graph x -axis ko cut kare).

Split middle term: $p + q = b$ AND $p \times q = a \times c$.

Linear $\rightarrow$ line, Quadratic $\rightarrow$ parabola.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: $3x^4 - 2x^2 + 7$ ka degree?

- Step: Saare terms ke power dekho: 4, 2, 0.
- Step: Sabse bada power = 4.

Answer: Degree = 4

Example 2: Kya $\sqrt{x} + 2$ polynomial hai?

- Step: $\sqrt{x} = x^{(1/2)}$, yani power $1/2$ (fraction).
- Step: Polynomial me fraction power allowed nahi.

Answer: Nahi

Example 3: $5x^2 - 3x + 8$ me x ka coefficient?

- Step: x (yani x^1) ke aage wala number dekho.
- Step: Wo -3 hai (minus ke saath).

Answer: -3

Example 4: $p(x) = x^2 - 4$, to $p(3) = ?$

- Step: x ki jagah 3 daal do: $3^2 - 4$.
- Step: $= 9 - 4$.

Answer: 5

Example 5: $p(x) = 2x - 6$ ka zero?

- Step: Zero ke liye $p(x) = 0$ rakho: $2x - 6 = 0$.
- Step: $2x = 6 \rightarrow x = 3$.

Answer: Zero = 3

Example 6: $4x^3 + 2x - 1$ ko classify karo.

- Step: Sabse bada power = 3.
- Step: Degree 3 wale polynomial ko CUBIC kehte hain.

Answer: Cubic

Example 7: $x^2 + 5x + 6$ ko factorise karo.

- Step: Yahan $a=1$, $b=5$, $c=6$. $a \times c = 6$.
- Step: $p+q = 5$ aur $p \times q = 6 \rightarrow p=2$, $q=3$.
- Step: $x^2 + 2x + 3x + 6 = x(x+2) + 3(x+2)$.

Answer: $(x + 2)(x + 3)$

Example 8: $x^2 - 5x + 6$ ke zeros?

- Step: Factorise: $(x-2)(x-3) = 0$.
- Step: $x - 2 = 0 \rightarrow x = 2$; $x - 3 = 0 \rightarrow x = 3$.

Answer: Zeros = 2 aur 3

Example 9: Kya 7 polynomial hai? Iska degree?

- Step: 7 ko $7x^0$ likh sakte hain (power 0).
- Step: Ye ek constant polynomial hai.

Answer: Haan, degree = 0

Example 10: $x^2 - 9$ ko factorise karo.

- Step: Ye $a^2 - b^2$ form hai: $x^2 - 3^2$.
- Step: Rule: $a^2 - b^2 = (a - b)(a + b)$.

Answer: $(x - 3)(x + 3)$

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1. $5x^3 - 2x + 1$ ka degree kya hai?
Q2. $3x^2 - 7x + 2$ me x^2 ka coefficient kya hai?
Q3. $4x + 9$ me constant kya hai?
Q4. Kya $x^{-2} + 3$ ek polynomial hai?
Q5. $p(x) = x^2 + 1$, to $p(0) = ?$
Q6. $p(x) = 2x - 8$ ka zero kya hai?
Q7. $ax^2 + bx + c$ kis type ka polynomial hai?
Q8. $6x$ ka degree kya hai?
Q9. 9 (sirf constant) ka degree kya hai?
Q10. $x^2 - 9$ ko factorise karo.
Q11. Quadratic polynomial me maximum kitne zeros hote hain?
Q12. $x^2 + 7x + 12$ ko factorise karo.
Q13. Linear polynomial ka graph kaisa hota hai?
Q14. Quadratic polynomial ka graph kya kehlata hai?
Q15. $p(x) = x^2 - 3x + 2$, to $p(1) = ?$
Q16. $5x^3 - 2x^2$ me x^3 ka coefficient kya hai?
Q17. Monomial, Binomial aur Trinomial kya hote hain?
Q18. $x^2 - 5x + 6$ ke zeros kya hain?
Q19. Splitting middle term me kaunsi 2 conditions chahiye?
Q20. Graph jahan x-axis ko cut kare, wo kya hota hai?

Step-by-Step Solutions

- Q1. $5x^3 - 2x + 1$ ka degree?
- Step: Sabse bada power dhoondo: 3, 1, 0.
- Step: Sabse bada = 3.
Answer: 3
- Q2. $3x^2 - 7x + 2$ me x^2 ka coefficient?
- Step: x^2 ke aage wala number = 3.
Answer: 3
- Q3. $4x + 9$ me constant?
- Step: Bina variable wala number = 9.
Answer: 9
- Q4. Kya $x^{-2} + 3$ polynomial hai?
- Step: Power -2 (negative) hai, jo allowed nahi.
Answer: Nahi
- Q5. $p(x) = x^2 + 1$, $p(0)$?
- Step: x ki jagah 0: $0^2 + 1 = 0 + 1$.

Answer:1

Q6. $p(x) = 2x - 8$ ka zero?

- Step: $2x - 8 = 0 \rightarrow 2x = 8 \rightarrow x = 4$.

Answer:4

Q7. $ax^2 + bx + c$ kis type ka?

- Step: Sabse bada power 2 hai.

Answer:Quadratic

Q8. $6x$ ka degree?

- Step: $6x = 6x^1$, power = 1.

Answer:1

Q9. 9 ka degree?

- Step: $9 = 9x^0$, power = 0.

Answer:0

Q10. $x^2 - 9$ factorise?

- Step: $x^2 - 3^2$ form hai.
- Step: $a^2 - b^2 = (a-b)(a+b)$.

Answer: $(x-3)(x+3)$

Q11. Quadratic me max zeros?

- Step: Degree 2 $\rightarrow$ maximum 2 zeros.

Answer:2

Q12. $x^2 + 7x + 12$ factorise?

- Step: $a \times c = 12$, $p+q=7$, $pxq=12 \rightarrow 3,4$.
- Step: $(x+3)(x+4)$.

Answer: $(x+3)(x+4)$

Q13. Linear ka graph?

- Step: Degree 1 ka graph seedhi line hoti hai.

Answer:Seedhi (straight) line

Q14. Quadratic ka graph?

- Step: Degree 2 ka graph U ya ulta U hota hai.

Answer:Parabola

Q15. $p(x) = x^2 - 3x + 2$, $p(1)$?

- Step: $x=1$: $1 - 3 + 2 = 0$.

Answer:0

Q16. $5x^3 - 2x^2$ me x^3 ka coefficient?

- Step: x^3 ke aage = 5.

Answer:5

Q17. Monomial / Binomial / Trinomial?

- Step: Terms ki ginti se: 1, 2, aur 3 terms.

Answer:1 term, 2 terms, 3 terms

Q18. $x^2 - 5x + 6$ ke zeros?

- Step: Factorise: $(x-2)(x-3) = 0$.
- Step: $x = 2$ ya $x = 3$.

Answer:2 aur 3

Q19. Splitting middle term ki 2 conditions?

- Step: Do numbers p , q chahiye.

Answer: $p + q = b$ AND $p \times q = a \times c$

Q20. Graph jahan x-axis ko cut kare wo?

- Step: Us point par $p(x) = 0$ hota hai.

Answer: Zero (root)